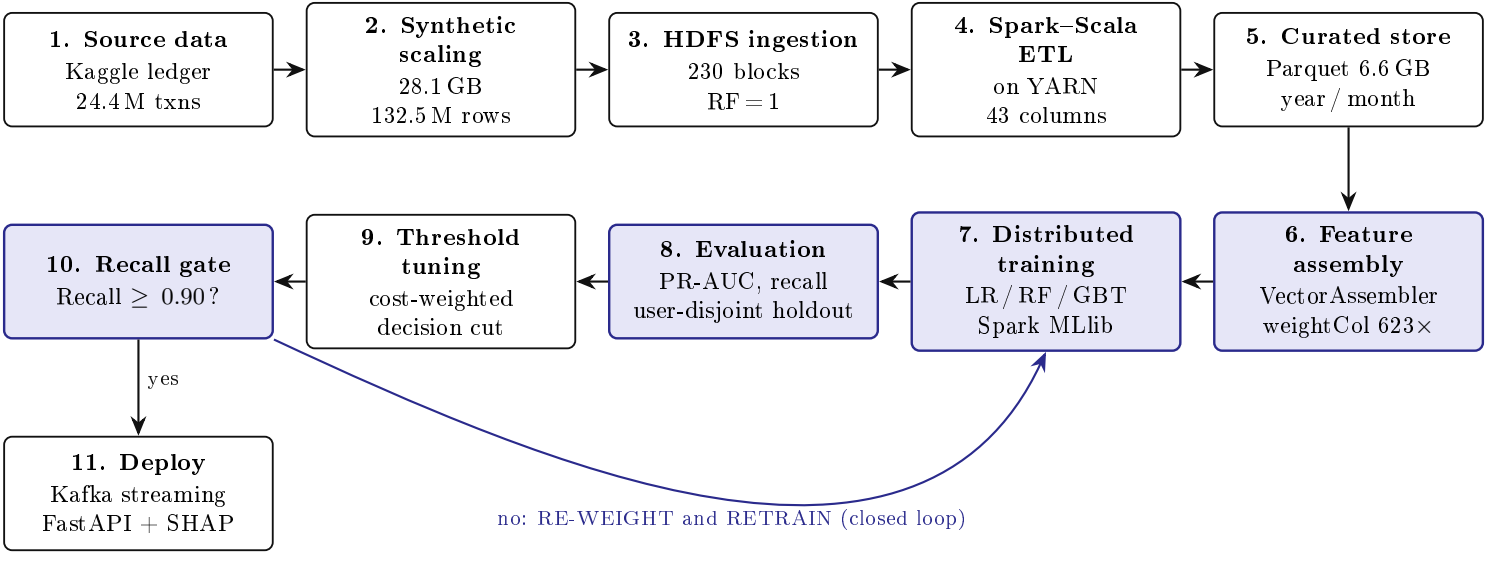

1. Source data
Kaggle ledger
24.4 M txns
2. Synthetic scaling
28.1 GB
132.5 M rows
3. HDFS ingestion
230 blocks
RF = 1
4. Spark–Scala ETL
on YARN
43 columns
5. Curated store
Parquet 6.6 GB
year / month
6. Feature assembly
VectorAssembler
weightCol 623×
7. Distributed training
LR / RF / GBT
Spark MLlib
8. Evaluation
PR-AUC, recall
user-disjoint holdout
9. Threshold tuning
cost-weighted
decision cut
10. Recall gate
Recall ≥ 0.90 ?
yes
11. Deploy
Kafka streaming
FastAPI + SHAP
no: RE-WEIGHT and RETRAIN (closed loop)